INCH-POUND

MIL-DTL-24633B
16 January 2026
SUPERSEDING
MIL-C-24633A
24 April 1990

DETAIL SPECIFICATION

CHAIN, ANCHOR, STUD LINK, FLASH BUTT WELDED, LOW ALLOY STEEL

Reactivated after 16 January 2026 and may be used for new and existing designs and acquisitions.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers ¾-inch through 4¾-inch, low alloy steel, flash butt welded, stud link, anchor chain for use on ships.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-24635 - Coating System, Weather-Resistant, Exterior Use

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-2035 - Nondestructive Testing Acceptance Criteria

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 4010

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NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9074-AS-GIB-010/271 - Requirements for Nondestructive Testing Methods

(Copies of this document are available online via Model Based Product Support (MBPS) at https://mbps.navsea.cloud.navy.mil/Windchill. To request an MBPS account, submit a ticket to the ServiceNow Assistance Portal (SNAP), https://snap.navair.navy.mil/mbps (DoD PKI Certificate required).)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM A370 - Standard Test Methods and Definitions for Mechanical Testing of Steel Products

ASTM E112 - Standard Test Methods for Determining Average Grain Size

ASTM E381 - Standard Method of Macroetch Testing Steel Bars, Billets, Blooms, and Forgings

ASTM E587 - Standard Practice for Ultrasonic Angle-Beam Contact Testing

(Copies of these documents are available online at www.astm.org.)

AMERICAN WELDING SOCIETY (AWS)

AWS A5.1/A5.1M - Specification for Carbon Steel Electrodes for Shielded Metal Arc Welding

AWS A5.5/A5.5M - Specification for Low-Alloy Steel Electrodes for Shielded Metal Arc Welding

AWS A5.20/A5.20M - Specification for Carbon Steel Electrodes for Flux Cored Arc Welding

AWS A5.26/A5.26M - Specification for Carbon and Low-Alloy Steel Electrodes for Electrogas Welding

AWS A5.28/A5.28M - Specification for Low-Alloy Steel Electrodes and Rods for Gas Shielded Arc Welding

AWS A5.29/A5.29M - Specification for Low-Alloy Steel Electrodes for Flux Cored Arc Welding

AWS B2.1/B2.1M - Specification for Welding Procedure and Performance Qualification

(Copies of these documents are available online at www.aws.org.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

NACE 3/SSPC-SP 6 - Commercial Blast Cleaning

(Copies of this document are available online at www.ampp.org.)

DET NORSKE VERITAS (DNV)

Certification Notes No. 2.6 - Certification of Offshore Mooring Chain

(Copies of this document may be obtained by mail from Det Norske Veritas, Post Office Box 300, 1322 Høvik, Norway.)

SAE INTERNATIONAL

SAE AMS-STD-595 - Colors used in Government Procurement

(Copies of this document are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Material. Steel used in the manufacture of chain links and studs shall be ingot-cast process or continuous-cast process with a minimum reduction in area of 12:1, fully killed, and made to fine-grain practice. Material composition and mechanical properties shall be traceable from steel production to final assembly with independent verification. Sufficient croppage shall be taken from the top and bottom of each ingot or beginning and end of each continuously cast strand to ensure removal of harmful defects or segregation that would adversely affect the chain quality. Bar stock made from material at the top of the ingot or end of the continuously cast strand after cropping shall be identified for testing. The bar stock for each heat of steel used in the manufacture of the links shall have an austenite grain size of 5 or finer in accordance with ASTM E112. Studs shall be made of steel of the same composition used in the links. Studs shall be forgings. Material used in normal production shall be the same composition of material used in the first article sample in chemistry, quality, and properties. The chain material shall be identified from steel mill to chain fabrication. Unidentified material shall be considered nonconforming. Bar stock for links may be selected greater than the nominal chain size (link diameter) according to table I, provided that the finished chain meets the dimensions and tolerances specified (see 3.3 and 4.4.2). Ovality of the bar stock shall be not greater than 0.75 times the bar stock tolerance.

TABLE I. Bar stock tolerances.

Chain size, inclusive (inches)	Permissible bar stock tolerance (inch)
¾ to 1¼	Plus 3/64
1⅜ to 2½	Plus 1/16
2⅝ to 3½	Plus 3/32
3⅝ to 4¾	Plus ⅛

3.2.1 Material chemical composition. The chemical composition of the steel shall be determined at the steel mill for each heat of steel and shall conform to table II (see 4.4.1.2). Alloying elements to achieve required strength and grain refinement shall be included, as necessary.

TABLE II. Chemical composition ladle analysis.

Element	Weight percent (maximum)
Carbon (C)	0.33
Silicon (Si)	0.37
Manganese (Mn)	1.94
Phosphorus (P)	0.035
Sulphur (S)	0.035

3.2.2 Material mechanical properties. The bar stock used in the manufacture of the chain links, after receiving identical heat treatment as the chain (see 3.15.4), shall possess the mechanical properties specified in table III when tested (see 4.4.1.3).

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TABLE III. Bar stock mechanical properties.

Property	Value (minimum)
Yield strength (0.2 percent offset)	68,000 lb/in^2
Ultimate tensile strength	99,600 lb/in^2
Elongation specimen gauge length – 4X specimen diameter	18 percent
Reduction in area	45 percent
Specimen impact, Charpy V-notch, average for three specimens at 32 °F	43 ft-lb

3.2.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Link weight and dimensions. Link weight and dimensions shall be in accordance with table IV, figure 1, and figure 2 (see 3.2). Chain link size (link diameter) shall be as specified (see 6.2). Link tolerances shall be in accordance with figure 2 and table V. Length over six links shall be measured from every third link (see 3.11, 4.4.10, and 4.5.1). A maximum flash trim overage permitted after the flash butt weld is deburred shall be as specified in table V (see 3.15.2.1). The grip radius of each end of the link shall be symmetrical to ensure proper fit and function in the chain handling equipment.

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TABLE IV. Stud link chain dimensions and properties.

Chain size (inches)	Link length A 1/ (inches)	Length over 6 links L 1/ (inches)			Link width B 1/ (inches)	Number of links per 15-fathom shot	Proof test load (lb)	Break test load (lb)	Min weight per 15-fathom shot (lb)	Max weight per 15-fathom shot (lb)
		Minimum	Nominal	Maximum						
¾	4½	19⅜	19½	19 13/16	2 11/16	359	48,000	75,000	465	525
⅞	5¼	22⅝	22¾	23 1/16	3⅛	305	64,400	98,000	640	713
1	6	25⅞	26	26⅜	3⅝	267	84,000	129,000	840	925
1⅛	6¾	29 1/16	29¼	29⅝	4 1/16	237	106,000	161,000	1,050	1,150
1¼	7½	32 5/16	32½	32 15/16	4½	213	130,000	198,000	1,310	1,430
1⅜	8¼	35 9/16	35¾	36¼	4 15/16	193	157,000	235,000	1,590	1,760
1½	9	38 13/16	39	39½	5⅜	177	185,000	280,000	1,890	2,080
1⅝	9¾	42	42¼	42⅞	5⅞	165	216,000	325,000	2,180	2,390
1¾	10½	45¼	45½	46⅛	6 5/16	153	249,000	380,000	2,520	2,750
1⅞	11¼	48½	48¾	49½	6¾	143	285,000	432,000	2,900	3,150
2	12	51 11/16	52	52¾	7 3/16	135	318,000	454,000	3,270	3,540
2⅛	12¾	54 15/16	55¼	56⅛	7⅝	125	357,000	510,000	3,690	3,980
2¼	13½	58 3/16	58½	59⅜	8⅛	119	396,000	570,000	4,150	4,450
2⅜	14¼	61 7/16	61¾	62⅝	8 9/16	113	440,000	628,000	4,640	4,960
2½	15	64 11/16	65	66	9	107	484,000	692,000	5,150	5,490
2⅝	15¾	67⅞	68¼	69¼	9 7/16	101	530,000	758,000	5,670	6,280
2¾	16½	71⅛	71½	72 9/16	9⅞	97	578,000	826,000	6,240	6,890
2⅞	17¼	74⅜	74¾	75⅞	10⅜	93	628,000	897,000	6,840	7,520
3	18	77⅝	78	79 3/16	10 13/16	89	679,000	970,000	7,460	8,180
3⅛	18¾	80 13/16	81¼	82½	11¼	87	732,000	1,046,000	8,090	8,890
3¼	19½	84 1/16	84½	85¾	11 11/16	83	787,000	1,124,000	8,760	9,600
3⅜	20¼	87 5/16	87¾	89	12⅛	79	843,000	1,204,000	9,470	10,350
3½	21	90 9/16	91	92 5/16	12⅝	77	900,000	1,285,000	10,220	11,140
3⅝	21¾	93 13/16	94¼	95⅝	12 15/16	73	958,000	1,369,000	10,990	12,190

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TABLE IV. Stud link chain dimensions and properties – Continued.

Chain size (inches)	Link length A [1/] (inches)	Length over 6 links L [1/] (inches)			Link width B [1/] (inches)	Number of links per 15-fathom shot	Proof test load (lb)	Break test load (lb)	Min weight per 15-fathom shot (lb)	Max weight per 15-fathom shot (lb)
		Minimum	Nominal	Maximum						
3¾	22½	97 1/16	97½	98⅞	13½	71	1,019,000	1,455,000	11,660	12,920
3⅞	23¼	100¼	100¾	102 3/16	14	69	1,080,000	1,543,000	12,540	13,850
4	24	103½	104	105½	14⅜	67	1,143,000	1,632,000	13,330	14,680
4⅛	24¾	106¾	107¼	108 13/16	14⅞	65	1,207,000	1,724,000	14,080	15,520
4¼	25½	110	110½	112⅛	15 5/16	63	1,272,000	1,817,000	14,830	16,360
4⅜	26¼	113 3/16	113¾	115⅜	15¾	61	1,338,000	1,911,000	15,570	17,200
4½	27	116 7/16	117	118 11/16	16 3/16	59	1,405,000	2,008,000	16,300	18,030
4⅝	27¾	119 11/16	120¼	122	16⅝	57	1,474,000	2,105,000	17,010	18,840
4¾	28½	122 15/16	123½	125 5/16	17⅛	57	1,700,000	2,550,000	18,370	20,300

NOTE:

[1/] Letters A, B, and L refer to the dimensions on figure 1.

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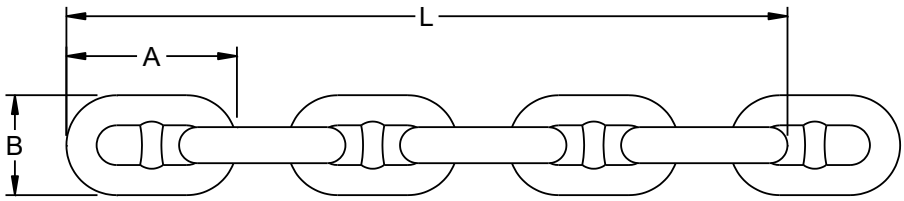


FIGURE 1. Stud link chain dimensions.

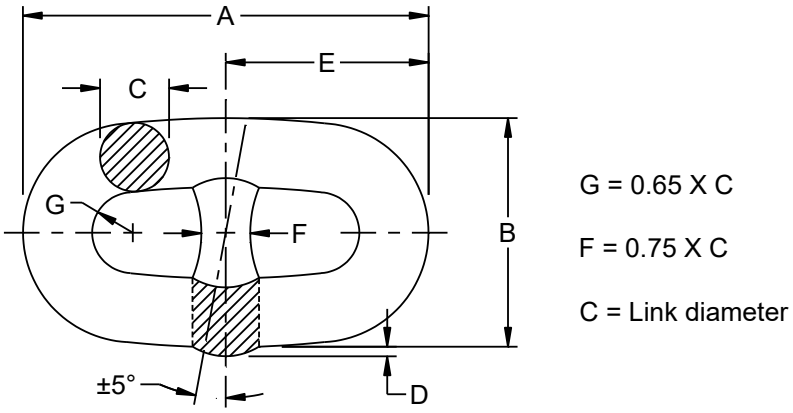


FIGURE 2. Basic link and stud dimensions.

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TABLE V. Basic link and stud tolerances.

Chain size, inclusive (link diameter) C [2/] (inches)	Permissible tolerance (inch)						Maximum flash trim overage D [2/] (inch)
	Link length A [2/] plus or minus	Link width B [2/] plus or minus	Link diameter ovality, maximum [1/]	Stud centering (E – A/2) [2/] plus or minus	Stud width F [2/] plus	Grip radius G [2/] plus or minus	
¾ to 1¼	1/16	1/16	1/32	3/16	1/16	1/16	1/16
1⅜ to 2½	3/32	1/8	3/64	5/16	3/32	3/32	1/16
2⅝ to 3	3/16	3/16	3/32	7/16	1/8	1/8	3/32
3⅛ to 3½	1/4	1/4	1/8	9/16	3/16	3/16	3/32
3⅝ to 4⅛	5/16	5/16	3/16	11/16	1/4	1/4	1/8
4¼ to 4¾	3/8	5/16	1/4	3/4	1/4	1/4	5/32

NOTES:

1/ When the link diameter (dimension C) is checked for ovality (major axis minus minor axis), the area of the elliptical cross section ($\pi/4 \times$ major axis $\times$ minor axis) shall be at least equal to the area of a circle of the chain size ($\pi/4 \times$ link diameter^2). Dimension C shall be measured at a point generally in accordance with figure 2.

2/ Letters A through G correspond to the dimensions on figure 2.

3.4 Link and stud surface. The link surface shall be free of cracks when examined in accordance with 4.4.3, 4.4.14.6, and 4.5.3. Burrs, irregularities, and rough edges shall be contour ground to a fair surface. Links and studs shall also be free of surface irregularities, dents, and undercutting in excess of the amounts shown in table VI when examined in accordance with 4.4.3 and 4.5.3.1. Care shall be given to inspection in the vicinity where the link is gripped by clamping dies during flash welding. The inside surface of the grip area of the chain shall be free of ridges, flashings, or other projections above the fair surface of the chain. Any ridges, flashings, or other projections in the grip area shall be ground flush with the fair surface of the chain link.

TABLE VI. Allowable surface defects.

Chain size, inclusive (inches)	Allowable surface defect (inch)
¾ to 2½	1/32
2⅝ to 3½	3/32
3⅝ to 4⅛	1/8
4¼ to 4¾	5/32

3.5 Flash butt weld surface. Link surface at the flash butt weld shall be free of cracks, lack of fusion, or gross porosity when examined in accordance with 4.4.4.

3.6 Stud size, placement, and alignment. Stud size and shape, alignment, and centering shall be in accordance with figure 2 (see 4.4.5).

3.6.1 Studs. The stud shall not have protrusions on either end of the saddle. The stud shall be pressed into the link body and shall have no protruding flash from the stud forging process. Stud penetration into the link body resulting from the stud pressing process shall be within the limits specified in table VII. Stud insertion shall be measured in accordance with 4.4.5.1. The perimeter of the stud saddle shall be radiused to prevent sharp entry into the link body.

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TABLE VII. Stud penetration.

Stud penetration at midplane		
Location	**Minimum 1/ (inch)**	**Maximum 1/ (inch)**
Unwelded side	0.01C	0.05C
Flash welded side	0.03C	0.07C

NOTE:

1/ Link diameter (see figure 2). The letter C refers to the dimension on figure 2.

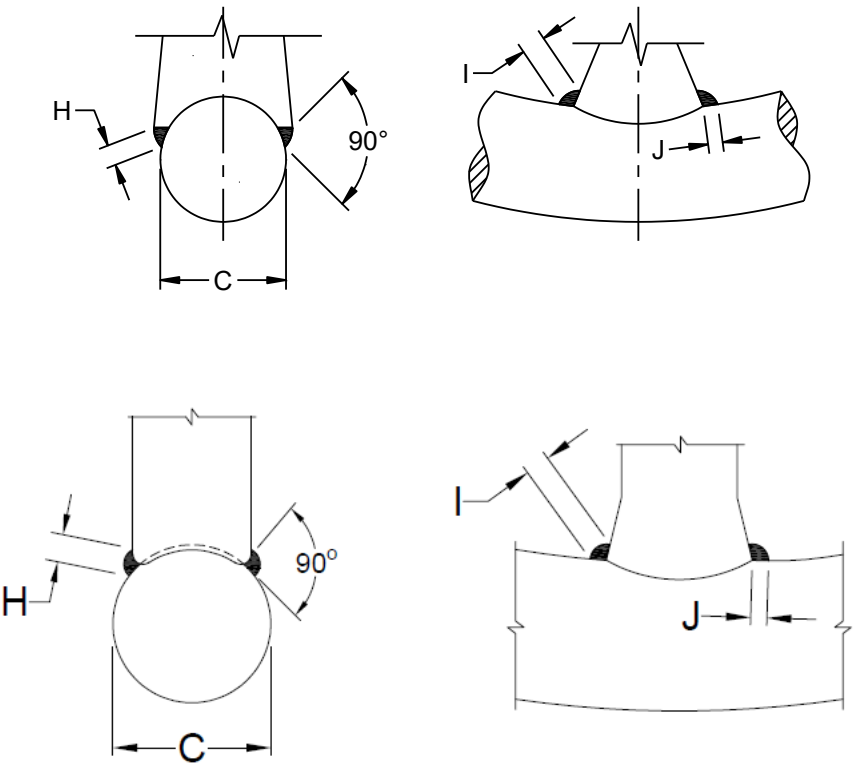
3.7 Flash butt weld fusion. The flash butt weld shall be free of defects causing ultrasonic back reflections equal to or greater than the standard when measured in accordance with 4.4.6. Ultrasonic testing inspection shall be in accordance with the Det Norske Veritas (DNV) Certification Note No 2.6 (see 4.4.6.1).

3.8 Link and stud interior. Links and studs shall be free from harmful defects such as laps, seams, pipes, cracks, fins, porosity, bursts, flakes, slivers, scabs, forged-in scale, nonmetallic inclusions, and undue segregation when examined in accordance with 4.4.7 and 4.5.14.

3.9 Fusion zone microstructure. The fusion zone of the flash butt weld and stud weld areas shall have fine grain structure (grain size 7 or finer in accordance with ASTM E112) and be free from Widmanstatten (needle-like) structure of any form when examined in accordance with 4.4.8, 4.4.8.1, and 4.4.8.2 after heat treating (see 3.15.4). The heat-affected zones on each side of the flash butt weld shall be examined in accordance with 4.4.8 and 4.4.8.1 for signs of overheating. The heat-affected zones shall be free of grain boundaries with intergranular fissuring and hot cracking. The stud weld fusion zones shall be free of cracks and fissures when examined in accordance with 4.4.8 and 4.4.8.2.

3.10 Stud weld. Magnetic particle inspection shall be performed on the stud weld (see 4.4.9), and the stud weld shall be free of cracks and fissures as specified in 4.5.12. Stud weld size shall be in accordance with figure 3.

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Dimension designation	Minimum [1/] dimensions (inches)
H [3/]	0.1C [2/]
I [3/, 4/]	0.2C
J [5/]	0.09C

NOTE:

1/ Link diameter (see figure 2).

2/ The letter C refers to the dimensions on figure 2.

3/ Dimensions H and I are weld face lengths.

4/ For chain link sizes of 2¼ and larger, the maximum dimension should be ½ inch.

5/ Dimension J is a weld leg length.

FIGURE 3. Stud weld dimensions.

3.11 Chain length. The chain shall be in 90-foot shots (15 fathoms), and the number of shots shall be as specified (see 6.2). A chain containing the number of links specified in table IV shall constitute a 90-foot shot. The overall length of six consecutive links measured from every third link shall be within the limits specified in table IV (see 3.3 and 4.4.10). By this method of measurement, every link, except three links at each end of the chain, shall be measured twice with different combinations of adjacent links totaling six links (see 4.5.1).

3.11.1 Chain weight. The weight of each shot of chain shall be within the limits specified in table IV and weighed in accordance with 4.4.10.1.

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3.12 Chain breaking load. The chain shall withstand the breaking load specified (see table IV and 4.4.11). There shall be no cracking in the flash butt weld, flash butt weld heat-affected zone, or in the area of the toe of the stud weld (see figure 3).

3.13 Chain proof load. The chain shall withstand the proof load specified (see table IV) without fracture (see 4.4.12). There shall be no cracking in the flash butt weld, flash butt weld heat-affected zone, or in the area of the toe of the stud weld (see figure 3).

3.13.1 Chain sizing. Chain may be sized (lengthened) to meet six-link length requirements by applying a load in excess of the proof load (see 4.4.12.1 and 4.5.1). However, such sizing shall not be done without the concurrence of the designated classification society surveyor (see 6.5). Sizing loads applied shall be recorded on strip charts.

3.13.2 Maximum sizing load. The sizing load shall not exceed 110 percent of the proof test load (see 3.13 and 4.4.12.2). Exceeding 110 percent of the proof test load shall be cause for rejection of the entire shot (where a full shot is sized in one operation) or the affected links (where fractions of a shot are sized in continuous production processes).

3.14 Chain mechanical properties. Links of samples detached from finished, heat-treated chain shall be sectioned for determination of mechanical properties (see 4.4.13 and 4.4.13.1). Mechanical properties shall be as specified in table VIII. In addition, the angle of bend shall be not less than 120 degrees before cracking or fracture (see 4.4.1.3.3).

TABLE VIII. Chain link mechanical properties.

Test	Value (minimum)
Yield strength (0.2 percent offset)	68,000 lb/in^2
Ultimate tensile strength	99,600 lb/in^2
Elongation (specimen gauge-length – 4X diameter)	16 percent
Reduction in area	40 percent
Charpy V-notch impact energy, unwelded portion of link, average for three specimens at 32 °F	43 ft-lb
Charpy V-notch impact energy, flash weld heat-affected zone (immediately adjacent to fusion line), average for three specimens at 32 °F	36 ft-lb

3.14.1 Link hardness. The link hardness, target hardness value for heat treatment, shall be established for a first article and conformance sample of chain in accordance with 4.4.13.1.

3.15 Standards of construction. Construction of the chain shall include the processes and equipment specified in 3.15.1 through 3.15.6. The material specification, process parameters, and equipment used to construct the first article inspection sample to this specification shall also be used in construction of production chain (see 4.4.14, 4.6, and 6.2).

3.15.1 Heating and bending. Bars shall be heated before bending using electric resistance heating (see 4.4.14.1). Peak bar temperatures shall not exceed 2,100 °F, and bending shall be performed at a bar temperature of 1,450 °F or greater. Reheating of the bar shall be permitted should the production sequence be delayed or interrupted. Alternate bar heating methods will be considered if acceptable documentation is presented for Government approval as specified (see 6.2).

3.15.1.1 Heating control. In order to prevent undue scaling or flaking of the bar stock due to excessive heat, the heating phase shall be automatically controlled by an optical heat sensor. The controller shall be checked for calibration at least once every 8 hours. Bar temperature and controller calibration records shall be available for inspection (see 4.4.14.1.1).

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3.15.2 Flash butt welding. The bent bar shall be electrically flash butt welded (see 4.4.14.2). Maximum bar end misalignment shall not exceed 5 percent of the bar diameter, and the bars shall be parallel within 5 degrees. Bar end misalignment shall be checked a minimum of once every 10 links after welding and systematically recorded.

3.15.2.1 Flash butt welding parameters. The following welding parameters shall be controlled during the welding of each link:

a. Platen motion.

b. Current as a function of time.

c. Hydraulic pressures.

These parameters, both in first article and normal production, shall be automatically recorded on strip charts (see 4.4.14.2.1). Excess flash butt weld material shall be removed to a uniform surface consistency over the entire link to the dimensional requirements of the link (see 3.3). Excess flash butt weld material shall be removed to a smooth contour with no depressions below the surface of the link body adjacent to the weld.

3.15.2.2 Flash removal inspection. A clean fusion zone free of all foreign material such as scale, dirt, and slag shall be maintained. A visual inspection of each link in the area to be covered by the stud shall be made after excess flash butt weld metal removal. Potentially harmful indications shall be tagged for later evaluation (see 4.4.14.2.2) or removal of the link.

3.15.3 Stud welding for particular chain sizes. For chain sizes 1½ inches and larger, the stud shall be completely welded, circumferentially, on the end opposite the flash butt weld using hydrogen controlled electrodes (EXX18) in accordance with AWS A5.1/A5.1M or AWS A5.5/A5.5M, or by metal inert gas or tungsten inert gas processes in accordance with AWS A5.26/A5.26M and AWS A5.28/A5.28M, or by flux-cored wire feed arc welding in accordance with AWS A5.20/A5.20M or AWS A5.29/A5.29M. The weld shall be of the size shown on figure 3 (see 4.4.14.3). Welding shall be performed using a welding preheat of 200 °F minimum to 300 °F maximum. Welding shall be performed prior to normalizing. The welding process and welders shall be qualified in accordance with AWS B2.1/B2.1M. For chain sizes smaller than 1½ inches, the studs shall be welded along the outside edges, parallel to the longitudinal direction of the link, on the end opposite the flash butt weld in accordance with the requirements previously stated in this paragraph.

3.15.4 Heat treatment. Chain shall be heat treated above the upper transformation temperature at a combination of temperatures and times to produce a fine-grain structure throughout the link body, flash and stud welds, and fusion zones of those welds (see 3.9, 4.4.8.1, 4.4.8.2, and 4.4.14.4). Heat treatment shall be performed after all welding has been completed and prior to testing and inspection. Chain shall be heat treated not more than twice. A system of identification shall be established to ensure that no shot, or portion thereof, is heat treated more than twice. Temperatures and times, or temperatures and chain speed, shall be controlled. Each link shall be maintained at a temperature between 1,500 °F minimum and 1,750 °F maximum for a period determined as a result of first article production. After exit from the furnace, the links shall be air-cooled, or quenched and tempered in such a manner that differential cooling does not occur relative to opposite sides of the links. A continuous-type furnace shall be used for normal production. Chain shall be removed from the furnace exit to prevent accumulating chain in areas with elevated temperatures that may affect the air-cooling.

3.15.5 Markings. The stud of each link shall be permanently marked, in raised letters or letters indented with a blunt nose tool, with “USN” and the chain size in fractional inches on one side, and with the brand name or trade name of the manufacturer on the other side. The studs of both end links of each 90-foot shot shall be marked with serial numbers assigned by the manufacturer and traceable to production, test inspection, and certification records (see 4.4.14.5 and 4.6). Markings shall be a minimum of $^{5}/_{32}$ inch in height for ¾- to 1⅛-inch chain, and ¼ inch in height for 1¼-inch chain and above. Markings shall be raised or indented a minimum of $^{1}/_{32}$ inch. An area for serial number marking may be prepared by grinding.

3.15.6 Finishing. The chain shall be commercially blasted clean at a minimum in accordance with NACE 3/SSPC-SP 6 to remove loose scale and welding slag. After satisfactory completion of all tests and inspections, the chain shall be coated with an application of anchor chain paint, one coat of MIL-PRF-24635 type V, class 2, grade B, color SAE AMS-STD-595/26270 followed by one coat of MIL-PRF-24635 type V, class 1, grade A color SAE AMS-STD-595/17038. The minimum total application thickness shall be at least 3 mils (see 4.4.14.6). Coating shall be dry to the touch (70 °F) after 24 hours. Heat used in finishing shall not exceed 150 °F.

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3.15.6.1 Surface quality. Prior to finishing the anchor chain, the surface shall be checked and defects shall be removed (see 4.4.14.6.1). The surface of links shall have a uniform surface finish (see 3.4).

3.16 Chain. Chain shall consist of all common links (see 4.4.15).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on anchor chains when a first article sample is required (see 3.1). This inspection shall include the examinations and tests specified in table IX. One first article sample successfully examined and tested for acceptability of characteristics as specified in 4.4 shall serve as a manufacturing standard and shall be used as a guide in the inspection of production anchor chains produced under the contract.

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TABLE IX. First article and conformance inspections.

Characteristic	First article	First article Sample	Conformance	Conformance Sample	Requirement	Verification
Material	X	4.2.1.4	X	4.3.1.1	3.2	4.4.1, 4.4.1.1
Material chemical composition	X	4.2.1.2	--	4.3.1.3	3.2.1	4.4.1.2
Material mechanical properties	X	4.2.1.2	X	4.3.1.5	3.2.2	4.4.1.3, 4.4.1.3.1, 4.4.1.3.2, 4.4.1.3.3
Link weight and dimensions	X	4.2.1.1 and 4.2.1.3	X	4.3.1.4	3.3	4.4.2
Link and stud surface	X	4.2.1.1	X	4.3.1.3	3.4	4.4.3, 4.4.14.6
Flash butt weld surface	X	4.2.1.1	X	4.3.1.3	3.5	4.4.4
Stud size, placement, and alignment	X	4.2.1.3	X	4.3.1.4 and 4.3.1.5	3.6	4.4.5
Studs	X	--	X	--	3.6.1	4.4.5.1
Flash butt weld fusion	X	4.2.1.1	X	4.3.1.3 and 4.3.1.5	3.7	4.4.6, 4.4.6.1
Link and stud interior	X	4.2.1.2	X	--	3.8	4.4.7
Fusion zone microstructure	X	4.2.1.2	X	4.3.1.5	3.9	4.4.8, 4.4.8.1, 4.4.8.2
Stud weld	X	4.2.1.1	X	4.3.1.3	3.10	4.4.9
Chain length	X	4.2.1.1	X	4.3.1.3	3.11	4.4.10
Chain weight	X	4.2.1.1	X	4.3.1.3	3.11.1	4.4.10.1
Chain breaking load	X	4.2.1.3	X	4.3.1.4	3.12	4.4.11
Chain proof load	X	4.2.1.1	X	4.3.1.3	3.13	4.4.12
Chain sizing	X	--	X	--	3.13.1	4.4.12.1
Maximum sizing load	X	--	X	--	3.13.2	4.4.12.2
Chain mechanical properties	X	--	X	--	3.14	4.4.13
Link hardness	X	4.2.1.2	--	--	3.14.1	4.4.13.1
Standards of construction	X	--	X	--	3.15	4.4.14
Heating and bending	X	--	X	--	3.15.1	4.4.14.1

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TABLE IX. First article and conformance inspections – Continued.

Characteristic	First article	First article Sample	Conformance	Conformance Sample	Requirement	Verification
Heating control	X	--	X	--	3.15.1.1	4.4.14.1.1
Flash butt welding	X	--	X	--	3.15.2	4.4.14.2
Flash butt welding parameters	X	--	X	--	3.15.2.1	4.4.14.2.1
Flash removal inspection	X	--	X	--	3.15.2.2	4.4.14.2.2
Stud welding for particular chain sizes	X	--	X	--	3.15.3	4.4.14.3
Heat treatment	X	--	X	--	3.15.4	4.4.14.4
Markings	X	--	X	--	3.15.5	4.4.14.5
Finishing	X	--	X	--	3.15.6	4.4.14.6
Surface quality	X	--	X	--	3.15.6.1	4.4.14.6.1
Chain	X	--	X	--	3.16	4.4.15

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4.2.1 First article inspection samples. As specified (see 6.2), the following samples shall be prepared for the first article inspection examinations and tests specified in table IX (see 4.2). Chain links shall be cleaned on all surfaces to remove scale. As a minimum, chain samples shall consist of the following.

4.2.1.1 Six-link section. One six-link section for the examinations of chain length (see 4.4.10), link surface (see 4.4.3), flash butt weld surface (see 4.4.4), flash butt weld fusion (see 4.4.6), stud weld (see 4.4.9), chain proof load test (see 4.4.12), and chain weight (see 4.4.2.1 and 4.4.10.1).

4.2.1.2 Finished links. Finished links, after heat treating as necessary, for the verification of chemical composition (see 4.4.1.2), the chain specimen tensile test (see 4.4.1.3.1), chain specimen impact test (see 4.4.1.3.2), chain specimen bend test (see 4.4.1.3.3), interior examination of flash weld (see 4.4.7), microstructure examination (see 4.4.8.1 and 4.4.8.2), and hardness test (see 4.4.13.1).

4.2.1.3 Triplet. One triplet, minimum, for the breaking load test (see 4.4.11) from a six-link section (see 4.2.1.1) with examination of link and stud dimensions (see 4.4.2) and stud alignment and centering and size (see 4.4.5).

4.2.1.4 Section of bar stock. One (8- to 12-inch) section of bar stock. The material shall be selected from bar stock of the same heat or melt of steel as the chain links used for first article inspection. The section of bar stock shall be used for bar stock mechanical properties tests (see 4.4.1.1.1 and 4.4.1.1.2) and the bar stock microstructure examination (see 4.4.1.1).

4.3 Conformance inspection. Bar stock and chain samples shall be selected in accordance with 4.3.1 and shall be inspected as specified in table IX. In addition, a detailed inspection of the chain shall be performed following cleaning of all surfaces to remove scale and after proof testing. The chain shall be arranged to permit careful inspection from all sides. The chain shall be free of paint or other coating that might conceal defects during the testing and inspection in accordance with T9074-AS-GIB-010/271.

4.3.1 Sampling for conformance inspection.

4.3.1.1 Bar stock material. For the purpose of sampling bar stock material for use in the construction of chain, one 8- to 12-inch-long section shall be selected from each heat or melt of steel. The section shall be removed from either end of any bar in the heat or melt. Bar stock mechanical properties tests (see 4.4.1.1.1 and 4.4.1.1.2) shall be conducted for each heat or melt of steel.

4.3.1.2 Finished chain. For the purpose of sampling finished chain, extra chain links shall be provided in each lot. A lot shall be defined as four shots plus additional links as described below, all from the same heat or melt of steel. Extra links shall be provided and processed with the shots of chain. All four shots in a lot shall be constructed with four extra links. The fourth link shall be cut to remove sample triplets.

4.3.1.3 Shots. The chain proof load test (see 4.4.12) and examination of chain length (see 4.4.10) shall be conducted on each shot of chain. Examinations of the link surface (see 4.4.3), flash butt weld surface (see 4.4.4), flash butt weld fusion (see 4.4.6), and stud weld (see 4.4.9) shall be conducted on each shot. The weight of finished shots shall be determined (see 4.4.10.1).

4.3.1.4 Detached triplets. A minimum of a triplet for each lot shall be removed for the chain breaking load test (see 4.4.11). Dimensions of each link of the triplet shall be measured for conformance to link dimensions (see 4.4.2), grip radius, and stud alignment and size (see 4.4.5) prior to the breaking load test.

4.3.1.5 Links. Extra links for each lot shall be removed in a manner that allows specimens to be taken for the chain specimen tensile test (see 4.4.1.3.1), chain specimen impact test (see 4.4.1.3.2), flash butt weld fusion zone microstructure examination (see 4.4.8.1), stud weld fusion zone microstructure examination (see 4.4.8.2), interior examination of flash butt weld fusion (see 4.4.6), and stud penetration examination (see 4.4.5.1).

4.3.1.6 Disposition of chain samples. Unless otherwise specified (see 6.2), all chain test samples and unused samples shall be shipped with the finished chain, retained at the manufacturing facility, or scrapped. If retention or shipping is desired, each sample shall be identified according to the shot from which it was taken.

4.4 Examinations and tests.

4.4.1 Material. Materials shall be visually inspected and tested to verify compliance with 3.2 and its subparagraphs.

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4.4.1.1 Bar stock. A sample of material shall be taken parallel to the bar long axis for examination of the microstructure with respect to the longitudinal axis of the bar (see 4.2.1.4). A ½-inch square area is the minimum size required for the bar stock microstructural examination and shall be taken from the center of the bar. The sample shall be polished and acid etched, and a microstructural examination shall be performed using 100× magnification (see 3.2).

4.4.1.1.1 Bar stock tensile test. The specimen shall be taken from the bar stock sample (see 4.2.1.4 and 4.3.1.1) and tested as specified (see 4.4.1.3.1) in accordance with ASTM A370. The specimen shall meet the requirements specified in 3.2.2.

4.4.1.1.2 Bar stock impact test. Three specimens shall be taken from the bar stock sample (see 4.2.1.4 and 4.3.1.1) and tested as specified (see 4.4.1.3.2) in accordance with ASTM A370. The specimens shall meet the requirements specified in 3.2.2.

4.4.1.2 Material chemical composition. The material sample shall be selected in accordance with 4.2.1.2 and shall be analyzed to determine the chemical composition. The bar stock chemical composition shall be in accordance with 3.2.1. The chemical analysis shall include added or limited alloying elements. The chemical composition (see 3.2.1) of the steel used in the manufacture of links and studs for first article testing shall be verified by a chemical or spectrographic analysis conducted independent of the steel supplier.

4.4.1.3 Material mechanical properties.

4.4.1.3.1 Chain specimen tensile test. The tensile specimen sample shall be taken from the link in the side opposite the flash butt weld (see 4.2.1.2 and 4.3.1.5). Orientation of the specimen shall be as shown on figure 4. Tensile tests shall be conducted with procedures and using standard specimens machined in accordance with ASTM A370 and shall meet the requirements specified in 3.14.

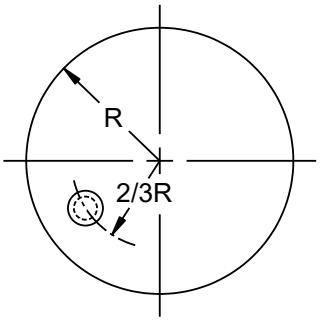


FIGURE 4. Orientation of tensile test specimen.

4.4.1.3.2 Chain specimen impact test. Six impact test specimens (see 4.2.1.2) shall be taken from one of the extra sample links (see 4.3.1.5). Three impact test specimens shall be taken across the weld, and three impact test specimens shall be taken across the unwelded side of the links. Orientation of the specimens shall be as shown on figure 5. The notch shall be centered in the flash butt weld heat-affected zone, ⅛ inch from the flash butt weld centerline, for the three specimens taken from that side of the link. Impact tests shall be conducted in accordance with ASTM A370 (see 3.14, 4.4.1.1.2, and 4.5.9). For chain sizes smaller than 1½ inches, the required impact test specimens may be taken from more than one link, and the notch orientation shall be maintained for the corresponding impact test specimens taken from the welded and unwelded sides of the same link.

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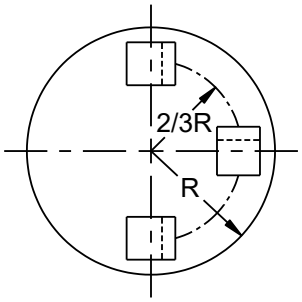
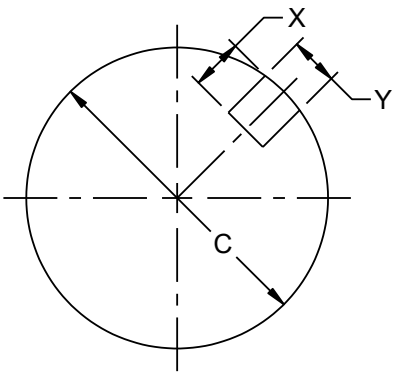


FIGURE 5. Orientation of Charpy V-notch impact test specimens.

4.4.1.3.3 Chain specimen bend test. The orientation of the flash butt weld bend test specimen (see 4.2.1.2) shall be as shown on figure 6. The sample shall be taken from an area of the link below the stud (see 4.2.1.2). The test piece shall be prepared with the maximum area of original bar surface retained and with a maximum thickness of 0.75 inch. The test shall be conducted with the bar surface in tension using a mandrel of the maximum radius shown in table X for the specimen thickness being tested. The sample shall meet the requirements specified in 3.14. The specimen shall be precisely positioned over the mandrel such that the weld area is parallel to and in contact with the mandrel.

TABLE X. Bend test mandrel radius.

Bend test specimen thickness (inch)	Mandrel maximum radius (inches)
1/4	11/16
5/16	7/8
3/8	1 1/16
7/16	1 1/4
1/2	1 1/2
9/16	1 5/8
5/8	1 13/16
11/16	2
3/4	2 1/4



X = 1/4-inch minimum to 3/4-inch maximum

Y = 1.5X or C

C = Link diameter (see Figure 2)

FIGURE 6. Orientation of bend test specimen.

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4.4.2 Link weight and dimensions. The link weight and dimensions (see 4.2.1.1 for sampling) shall be visually inspected and tested to verify compliance with 3.3. The dimensions of each link, including grip radii, of each detached triplet (see 4.3.1.4) shall be measured. Dimensions in table IV shall be checked when dimensional discrepancies are observed in visual examination of each shot (see 4.3), and as required for links of triplets detached from finished chain (see 4.5.13).

4.4.2.1 First article. For the first article inspection, the sample six-link section shall be weighed (see 4.2.1.1). The minimum and maximum weight of the six-link section shall be calculated from the information listed in table IV. The minimum and maximum shot weights shall be divided by the number of links per shot and the results multiplied by 6 to establish the minimum and maximum weights for the six-link section.

4.4.3 Link and stud surface. The link and stud surface shall be visually inspected to verify compliance with 3.4. After cleaning all surfaces to remove scale and after proof testing, and before painting, all surfaces of each link and stud shall be visually examined. For first article testing, a six-link section shall be used (see 4.2.1.1).

4.4.4 Flash butt weld surface. The flash butt weld surface shall be visually inspected to verify compliance with 3.5. Magnetic particle or liquid penetrant inspection means shall be employed to examine the flash butt weld surface of the link. Procedures applied for magnetic particle examination and liquid penetrant inspection shall be in accordance with T9074-AS-GIB-010/271. Acceptance criteria shall be in accordance with MIL-STD-2035. This examination shall be done after cleaning all surfaces to remove scale and after proof testing and before painting. Frequency of examination shall be in accordance with table XI. For first article testing, a six-link section shall be used (see 4.2.1.1).

TABLE XI. Frequency of examination of flash butt weld surface, flash butt weld fusion, and stud weld.

Chain size, inclusive (inches)	Frequency of examinations
¾ to 1	End links and every tenth link
1⅛ to 1⅜	End links and every eighth link
1½ to 1⅞	End links and every sixth link
2 to 2¾	End links and every fourth link
2⅞ to 4⅝	End links and every second link
4¾	All links

4.4.5 Stud size, placement, and alignment. The stud size, placement, and alignment shall be visually inspected to verify compliance with 3.6. The stud alignment and centering and the size of each circumferential stud weld shall be measured on the stud of each link of each detached triplet (see 4.5.13).

4.4.5.1 Studs. The stud penetration, protrusion, and insertion and the stud saddle shall be visually inspected and tested to verify compliance to 3.6.1.

4.4.6 Flash butt weld fusion. Flash butt weld fusions shall be tested to verify compliance to 3.7. Ultrasonic examination in accordance with the DNV Certification Note No 2.6 shall be employed to examine the flash butt welded fusion area of links. Procedure for ultrasonic inspections shall be in accordance with T9074-AS-GIB-010/271. The frequency of examination shall be in accordance with table XI.

4.4.6.1 Ultrasonic examination. Ultrasonic testing inspection shall be in accordance with DNV Certification Note No 2.6 as follows. Ultrasonic testing shall be performed on all links to test the entire flash weld area for proper fusion and upset. Testing shall be performed in accordance with ASTM E587 or another recognized standard using angle-beam shear waves in the range from 45 to 70 degrees as appropriate. Equipment shall be calibrated using International Institute of Welding calibration blocks. The search unit shall be checked for beam exit point and angle of reflection at least once per working shift or 8 hours, whichever comes first. Reference blocks shall be made from a chain link that is similar in diameter, surface condition, chemistry, and processing history to the production links being tested. The block shall contain two reference reflectors in the form of surface notches in the plane of the weld oriented 180 degrees apart; one located on the inner surface adjacent to the stud, and one located on the outer surface. The notch shall be maximum 3 millimeters wide and cut to a depth 4 percent of the nominal diameter or

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5mm, whichever is smaller. The notch shall be cut circular with radius 15mm. With the search unit positioned as appropriate, calibrate the instrument to obtain an indication amplitude from both reflectors of approximately 75 percent full screen height. Repeat the procedure from the other size of the weld. Perform the testing by scanning from both sides of the weld with the amplitude calibration increased by 6 dB. A link containing indications equal to or larger in amplitude to that of the reference notch, when properly corrected for distance, is subject to rejection, or repair if applicable.

4.4.7 Link and stud interior (macrostructure). The link and stud interior shall be visually inspected and tested to verify compliance to 3.8. A single link shall be cut in half on a plane parallel with both the long and short axes, etched, and examined (see 4.5.14). The sample shall be one piece or in three pieces from the same link, approximately equal in size, cut transverse to the long axis, and consisting of the weld and stud section and both bend sections. ASTM E381 shall be used to rate the significance of indications observed in the macrostructure. The insertion penetration of the stud into the link body as a consequence of the stud pressing operation shall be determined and conform to 3.6.1, 3.8, and table VII.

4.4.8 Fusion zone microstructure. The flash butt weld and stud weld fusion zone microstructure shall be tested to 4.4.8.1 for flash butt welds and 4.4.8.2 for stud welds to verify conformance to 3.9.

4.4.8.1 Flash butt weld fusion zone microstructure. A sample of material shall be taken parallel with the length of the link, traversing the flash butt weld fusion zone, and extending inward from the outside surface to the center of the bar (see 4.3.1.5). A 1-inch square area is considered the minimum size for a microstructure inspection for chain 1½ inches and over. For smaller sizes, the largest practicable sample shall be used. The flash butt weld zone shall be polished and acid etched. A microstructural inspection shall be performed using 100× magnification for conformance to 3.9 and 3.15.4. For first article testing, a six-link section shall be used (see 4.2.1.1).

4.4.8.2 Stud weld fusion zone microstructure. A sample of material shall be taken parallel with the length of the link, which includes the stud weld fusion zone, including the region where the stud and link body intersect (see 4.3.1.5). A 1-inch square area is the minimum size for a microstructure inspection for chain 1½ inches and over. For smaller sizes, the largest practicable sample shall be used. The fusion zone shall be polished and acid etched, and a microstructural inspection shall be performed using 100× magnification for conformance to 3.9 and 3.15.4.

4.4.9 Stud weld. The stud welds shall be tested to verify conformance to 3.10 and 3.15.3. Stud welds shall be completely magnetic particle inspected about the entire circumferential weld to 3.15.3. This examination shall be done after cleaning all surfaces to remove scale and after proof testing, and before painting. Magnetic particle inspection procedures and equipment shall be in accordance with T9074-AS-GIB-010/271 and acceptance criteria in accordance with MIL-STD-2035. Frequency of examination shall be in accordance with table XI. All stud welds of first article sample chain shall be inspected. For first article testing, a six-link section shall be used (see 4.2.1.1).

4.4.10 Chain length. The chain length shall be visually inspected to verify compliance to 3.11. Chain length measurements shall be made after proof testing with the chain links under a tension of 10 percent of proof load, and in a position to ensure that links are aligned and in full contact. For first article testing, a six-link section shall be used (see 4.4.2.1). The overall length of six links shall be measured from every third link of each shot for conformance to table IV (see 3.3, 3.11, and 4.5.1).

4.4.10.1 Chain weight. Chain weight shall be tested to verify compliance with 3.11.1. Finished chain shall be weighed on completion of marking (see 3.15.5) and finishing (see 3.15.6). For first article testing, a six-link section shall be used (see 4.4.2.1).

4.4.11 Chain breaking load test. The chain breaking load shall be tested to verify compliance to 3.12. Samples for break tests shall be tested by securing them in a testing machine in such a manner that the chain shall be free from twist; and the holding arrangement shall be such that all stresses bearing on the end links of the section under test shall be the same as those applied to every link tested. If, in normal production only, it is considered that the test equipment would be endangered by a sudden break, it shall be considered acceptable if the triplet is loaded to the required breaking strength and maintained at the load for 15 seconds. For first article testing, a six-link section shall be used. Samples in first article testing shall be tested to destruction. The test triplets shall be disposed of in accordance with 4.3.1.6. The break testing machine shall have been calibrated with traceable certificates within the 12 months prior to the chain breaking load test (see 4.5.5 and 4.5.6).

4.4.12 Chain proof load test. The chain proof load shall be tested to verify compliance to 3.13. Shots and first article sample chain shall be tested by securing them in a testing machine in such a manner that the chain shall be

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free from twist; and the holding arrangement shall be such that all stresses bearing on the end links of the section under test are the same as those applied to every link tested. The proof load testing machine shall have been calibrated with traceable certificates within the 12 months prior to the chain proof load test. Chain to be sized shall not be loaded in excess of 110 percent of the proof load (see 3.13.1). The chain shall meet the dimensional tolerance of table V post testing. Strip chart recordings of the applied load shall be taken of all proof load tests during both first article and conformance inspection tests.

4.4.12.1 Chain sizing. The chain size shall be visually inspected to verify compliance to 3.13.1.

4.4.12.2 Maximum sizing load. The maximum size loading shall be tested in accordance with 4.4.12 to verify compliance to 3.13.2.

4.4.13 Chain mechanical properties. The mechanical properties of the chain shall be tested to verify compliance to 3.14.

4.4.13.1 Link hardness. The link hardness shall be tested to verify compliance to 3.14.1. The hardness of a test link (see 4.2.1.2) shall be established on the outside of the link on the side opposite the flash butt weld in accordance with ASTM A370. A minimum of three readings is required.

4.4.14 Standards of construction. The chain construction shall be visually inspected and tested to verify compliance with 3.15 and its subparagraphs.

4.4.14.1 Heating and bending. Bars shall be heated and bent to verify compliance to 3.15.1.

4.4.14.1.1 Heating control. The heating control shall be tested to verify compliance to 3.15.1.1.

4.4.14.2 Flash butt welding. The flash butt welds shall be visually inspected and tested to verify compliance to 3.15.2. Ultrasonic examination in accordance with the Det Norske Veritas (DNV) Certification Note No 2.6 and 4.4.6.1 shall be employed to examine the flash butt welded fusion area of links. Procedure for ultrasonic inspections shall be in accordance with T9074-AS-GIB-010/271.

4.4.14.2.1 Flash butt welding parameters. The flash butt welding parameters shall be visually inspected and tested to verify compliance to 3.15.2.1.

4.4.14.2.2 Flash removal inspection. A clean fusion zone shall be visually inspected to verify compliance to 3.15.2.2.

4.4.14.3 Stud welding for particular chain sizes. All chain sizes shall be visually inspected. For chain sizes 1½ inches or larger, the stud welds shall be visually inspected and tested to verify compliance to 3.15.3.

4.4.14.4 Heat treatment. Proper heat treatment shall be tested in accordance with 4.4.14.1 to verify compliance with 3.15.4.

4.4.14.5 Markings. The markings shall be visually inspected to verify compliance with 3.15.5.

4.4.14.6 Finishing. The chain finish shall be visually inspected to verify compliance with 3.4 and 3.15.6.

4.4.14.6.1 Surface quality. The surface shall be inspected to verify compliance with 3.15.6.1. Allowable surface defects shall be in accordance with table VI. Defects outside of the allowed values shall be removed and the surface blended to get a smooth contour. The cross-sectional area in way of grinding shall be not less than the theoretical area of the nominal chain diameter. Links repaired by grinding shall be subject to non-destructive testing.

4.4.15 Chain. Chain links shall be visually inspected to verify compliance with 3.16.

4.5 Retest and rejection criteria.

4.5.1 Length. If this length of six consecutive links is short, the chain may be stretched by loading above the proof test load specified in table IV, providing this load does not exceed the proof load by more than 10 percent (see 3.13.1 and 4.4.12). If the chain length over six links exceeds the tolerances shown on figure 2, the overlength chain links shall be cut out and 4.5.2 shall apply (see 4.4.2), except first article sample chain, which shall conform to the dimensions shown in table IV.

4.5.2 Single links. If single links are found to be defective or to not meet other applicable requirements, defective links may be cut out and a connecting link inserted in their place. After the entire shot has successfully

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completed all manufacturing and inspection, the connecting link shall be removed, and the two resulting lengths may be used to start new shots of chain. No shot or portion thereof shall be heat treated more than twice.

4.5.3 Cracks. If a crack is found, it shall be ground down not more than 1/16 inch and streamlined to provide no sharp contours. The link shall then be rechecked for conformance to 3.4; otherwise, 4.5.2 shall apply. If flashings are found in the grip area, they shall be ground down in accordance with 3.4 and rechecked; otherwise, 4.5.2 shall apply.

4.5.3.1 Surface irregularities. If a surface irregularity, dent, or undercut exceeding the criteria of 3.4 in depth is found, this shall be cause for rejection of the link and 4.5.2 shall apply.

4.5.4 Flash butt weld surface. In flash butt weld surface examination by dye penetrant or magnetic particle inspection, the area shall be free of linear indications greater than 1/16 inch. If a defect in the flash butt weld is found, the area shall be ground down not more than 1/16 inch and streamlined to provide no sharp contours. The repaired area shall be examined by dye penetrant or magnetic particle inspection (see 4.4.4), and if a defect is still present, 4.5.2 shall apply.

4.5.5 Fracture in flash butt weld. If a crack or fracture occurs in the flash butt weld or flash butt weld heat-affected zone, the connected shot shall be rejected. In addition, another chain breaking load test triplet representative of the remaining three shots constructed in the same lot shall be subjected to the breaking load test. If the triplet fractures in the flash butt weld or flash weld heat-affected zone, the lot shall be rejected (see 4.4.11).

4.5.6 Break. If the chain breaks below the level specified in table IV by any mode, the connected shot shall be rejected. In addition, another chain breaking load test triplet representative of the remaining three shots constructed in the same lot shall be subjected to the breaking load test. If this triplet breaks below the level specified in table IV by any mode, the lot shall be rejected (see 4.4.11).

4.5.7 Proof test failure. Links subject to the proof test shall be examined accordance with 4.4.3, 4.4.4, and 4.4.6. If there is sign of fracture, including breaking or cracking, during the proof test, 4.5.2 shall apply.

4.5.8 Tensile test failure. If the tensile test fails to meet the requirements in table III or table VIII, but is within 2,000 pounds per square inch (lb/in^2) of the required tensile strength or within 2 percent of the required elongation, a retest of another specimen selected from the same sample is permissible. If the second test fails to meet the requirements of table III or table VIII, the lot of four shots or bar stock (as applicable) shall be rejected (see 4.4.1.1.1 and 4.4.1.3.1)

4.5.9 Impact test failures. The average results of the three impact tests shall meet or exceed the minimum requirements specified in table III or table VIII. If the average fails to meet the minimum requirement by an amount not exceeding 5 foot-pounds (ft-lb) or the value for more than one specimen fails to meet the requirement, or when the value for only one specimen is below 70 percent of the requirement, three additional specimens from the same samples may be tested and the results added to those previously obtained to form a new average. If the new average fails to meet the requirements of table III or table VIII, or if more than two individual results are lower than the requirements of table III or table VIII, or more than one result is below 70 percent of the requirements, the lot of four shots or bar stock (as applicable) shall be rejected (see 4.4.1.1.2 and 4.4.1.3.2).

4.5.10 Bend test failure. If there is cracking or fracture associated with the flash butt weld or in the heat-affected zone in the bend specimen before 120 degrees of bend is made, the lot shall be rejected (see 4.4.1.3.3).

4.5.11 Ultrasonic inspection failure. If indications of interior flash butt weld fusion defects in reference to the accepted calibration standards are detected during ultrasonic examination of the flash weld zone, 4.5.2 shall apply (see 4.4.6 and 4.4.6.1).

4.5.12 Magnetic particle inspection failure. In stud weld examination by magnetic particle inspection (see 3.10, 4.4.8.2, and 4.4.9), all welds and at least ½ inch of adjacent link body shall be free of linear indications greater than 1/16 inch. If a crack or indication is found, the area shall be ground down not more than 1/16 inch and streamlined to ensure no sharp contours and retested. If the crack or indication is still present, all the studs in the shot shall be inspected and rewelded (if necessary). The shot shall be normalized if repair welding has occurred, and the stud examination performed a second time. Failure of any stud welds during the second examination shall be cause for rejection of that shot. No shot, or portion thereof, shall be normalized more than twice.

4.5.13 Dimensions. If link diameter, length, width, grip radii, or stud alignment and centering, or stud or stud weld dimensions do not conform to the required dimensions, these shall be compared to the dimensions of other

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links in the triplet and links of all sample chains representing the lot. If a single particular dimension fails to meet the required dimensional tolerance in more than one-half of the sample links, all links in each shot of the lot shall be examined for link dimensions, grip radii, stud alignment, and size. If a single particular dimension fails to meet required dimensions in five of the links in a shot, the shot shall be rejected, and 4.5.2 shall apply (see 3.6, 4.4.2, and 4.4.5).

4.5.14 Macrostructure and microstructure. Rejection in macrostructural and microstructural examinations shall be for defects, cracks, or discrepancies in regard to grain size or microstructure requirements (see 3.8, 3.9, 4.4.7, 4.4.8.1, and 4.4.8.2). If stud penetration exceeds the amount specified in table VII, the other two detached links in the shot shall be checked. If one other link shows excessive stud penetration, the lot shall be rejected.

4.5.15 Material composition. Material not within the composition limits shall be rejected (see 4.4.1.2).

4.5.16 Chain weight. Shots of chain not within the weight limits of table IV shall be rejected.

4.6 Manufacturing process documentation. The process used to manufacture the chain assembly components in accordance with specified material properties shall be documented (see 6.2).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The chain covered by this specification is intended to be used for ship anchoring. Current stock of flash butt welded chain may be used until depleted; future requisitions for flash butt welded chain should refer to the current version of this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When first article inspection is required and the samples used in the first article inspection to be made available to the Government (see 3.1, 4.2.1, and 6.4).

c. Number of shots of chain required (see 3.11).

d. Chain size (link diameter) (see 3.13.1).

e. Requirements for Government approval of alternative bar heating methods (see 3.15.1).

f. Disposition of chain samples if other than specified (see 4.3.1.6).

g. Requirement to provide manufacturing process documentation (see 4.6 and 6.3.1).

h. Packaging requirements (see 5.1).

i. Requirements for classification society inspection if other than specified (see 6.5).

6.3 Certification. To be considered a certified producer of anchor chain and eligible to bid on solicitations to this specification, a contractor should currently be certified by the American Bureau of Shipping to produce extra-high strength, grade 3a flash butt welded stud link anchor chain, and authorized to use the American Petroleum Institute (API) monogram on chain manufactured in accordance with API SPEC 2F. A contractor bidding on solicitations to this specification must identify the place of manufacture of the chain, the source of rolled bars, and the source of forged studs to be used in fulfilling any resulting contract.

6.3.1 Retention of records. Agencies using this specification to acquire anchor chain for use on U.S. Navy ships should forward a copy of all data to Naval Sea Systems Command, Deck and Replenishment Systems Division (SEA 05Z43), Department of the Navy, Washington, DC 20362-5101 for retention.

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6.4 First article. When a first article inspection is required (see 6.2), the items should be a first article sample. The first article should consist of the number of units specified. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product that has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Classification society inspection. Unless otherwise specified, the contractor should acquire the services of a surveyor primarily from a member society of the International Association of Classification Societies during the manufacture and testing of chain made to this specification. The qualifications and experience of the particular surveyor (by name and society) should be submitted to and approved by the Government before the first article construction begins. The contractor should maintain an inspection and record-keeping system satisfactory to the surveying society. The surveyor should witness the measurement and testing of all chain.

6.6 Subject term (key word) listing.

Anchor chain
Chain proof load
Hardness
Stud welding

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Navy – SH
Air Force – 71

Preparing activity:
Navy – SH
(Project 4010-2025-001)

Review activities:
Navy – CG, YD
DLA – GS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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